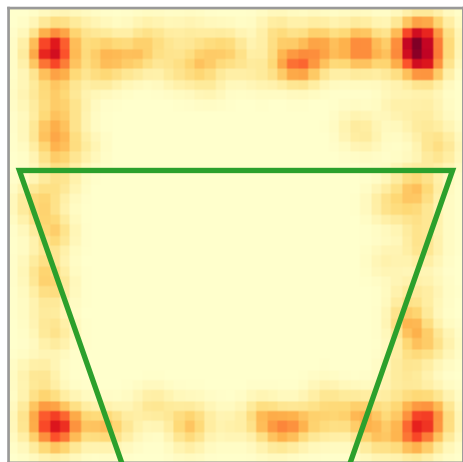


Closed-loop environment evolution: the skill-library signal moves the training distribution into the learnable band, while single-policy deploy success stays saturated ($p \approx 1$) and cannot drive this change.

Current environment e_t (round 5)

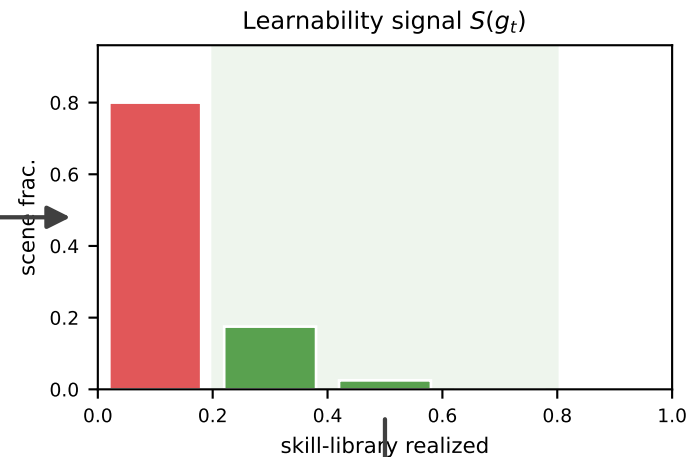


skill lib. realized=0.16 | deploy $p=0.95$

train policy
 π_{t+1} , iterate

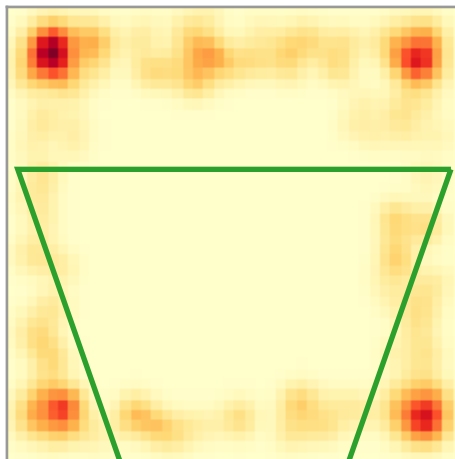
probe with
skill library
 $\pi, w_1 \dots w_K$

realized
distribution



LLM rewrites
generator program G

Evolved environment e_{t+1} (round 6)



skill lib. realized=0.35 (in band) | deploy $p=1.00$

Direction from $S(g_t)$: most scenes still too hard for the library (realized < 0.2 : 80%).
 $\Rightarrow$ rewrite generator to keep scenes solvable and move mass toward realized ≈ 0.5 (the learnable band), not merely harder.